Parent's signature

Name	Index Number	CTG

YISHUN JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2018

BIOLOGY **9744/03**

HIGHER 2 **12 SEPTEMBER 2018**

Paper 3 Long Structured and Free-response Questions **Wed 0800 - 1000**

2 hours

Additional Materials: Writing papers

READ THESE INSTRUCTIONS FIRST

Write your name and CTG in the spaces at the top of this page and on all separate answer paper used.
Write in dark blue or black pen only.
You may use a soft pencil for any diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question on the separate answer papers provided.

The use of an approved scientific calculator is expected, where Appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE	
Section A	
Q1	/30
Q2	/10
Q3	/ 10
Section B	
Q4 / Q5	/25
Overall	/ 75

This document consists of **15** printed pages and **3** blank pages.

Section A

Answer **all** questions.

1 The Human Immunodeficiency Virus (HIV) infects helper T lymphocytes, including memory helper T lymphocytes, thereby weakening the immune system, causing the patient to succumb to secondary infections, leading to the development of acquired immunodeficiency diseases (AIDS).

Fig. 1.1 shows the binding of a HIV particle to a T lymphocyte. The viral receptor gp120 trimer binds to both CD4 and a co-receptor called CCR5 on the T lymphocyte surface.

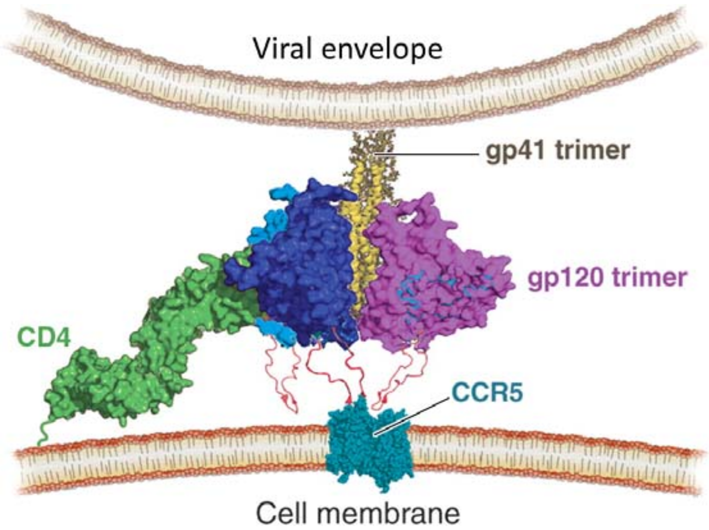


Fig. 1.1

(a) Individuals who carry a mutation known as CCR5–Δ32 in both copies of the CCR5 gene are resistant to certain strains of HIV. The CCR5–Δ32 allele is a result of a deletion of a particular sequence of 32 base-pairs within the exon of CCR5 gene.

Explain how an individual who is homozygous for the CCR5–Δ32 allele is resistant to HIV.

....

....

....

....

....

....

....

....[4]

BLANK PAGE

(b) The allele frequency of CCR5-Δ32 is unusually high in the western European countries, with up to 20% of ethnic western Europeans carrying this allele, which is rare or absent in all other ethnic groups throughout the world.

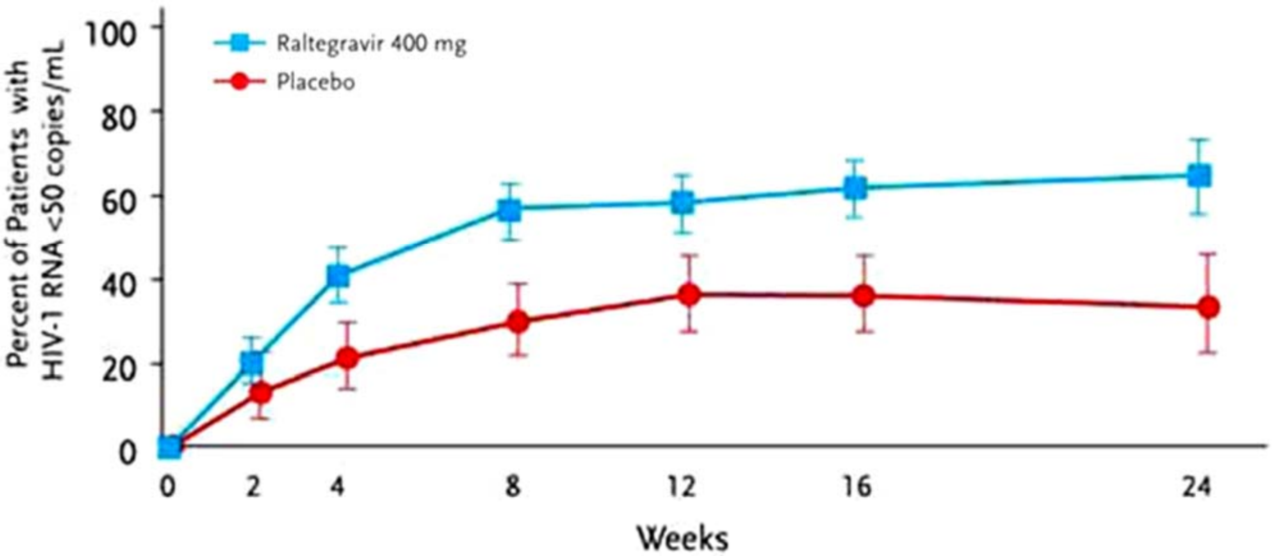
Suggest a reason for this phenomenon.

……………………………………………………………………………………………………….

……………………………………………………………………………………………………….[1]

There are two types of HIV: HIV-1 and HIV-2. HIV-1 integrase binds to specific sites at the ends of the viral DNA to facilitate its integration into the host genome. This allows the virus to persist in the latent stage for long periods of time.

Many integrase inhibitors have been discovered during the past 10 years, with some of them presently in clinical trials. An example of one such inhibitor is Raltegravir. In one such trial, one group of patients was given Raltegravir while the control group was given placebos (drug-free pills) over 24 weeks. The number of copies of HIV-1 RNA per ml of blood plasma was measured and the results are show in Fig. 1.2.



Adapted from Merck & Co., Inc. Results of BENCHMRK-1 and-2, two phase-III studies evaluating the efficacy and safety of raltegravir, a novel HIV-1 integrase inhibitor, in patients with triple-class resistant virus. 14th Conference on Retroviruses and Opportunistic Infections; February 25–28, 2007; Los Angeles, CA.

Fig. 1.2

(c) With reference to Fig. 1.2, describe the effect of treating patients with 400mg of Raltegravir.

……………………………………………………………………………………………………….

……………………………………………………………………………………………………….

……………………………………………………………………………………………………….

…………………………………………………………………………………………………….[2]

The onset of AIDS, which can occur many years later, coincides with a severely lowered primary and secondary immune response, owing to greatly reduced numbers of helper T lymphocytes in the body.

(d) Explain how the immune system defends against HIV infection.

……………………………………………………………………………………………………….

……………………………………………………………………………………………………….

……………………………………………………………………………………………………….

……………………………………………………………………………………………………….

……………………………………………………………………………………………………….

……………………………………………………………………………………………………….

……………………………………………………………………………………………………….

…………………………………………………………………………………………………….[4]

(e) Most T lymphocyte development occurs in an organ called the thymus. As part of this development, DNA sequences are cut out from the T lymphocyte genes, forming T lymphocyte receptor excision circles (TRECs).

The number of TRECs is thought to correlate with thymus function. CD4+ progenitor cells from individuals progressing to AIDS have a marked loss in T lymphocyte development capacity. The presence of TRECs in T lymphocytes can be detected in blood samples of HIV-infected patients using PCR.

(i) Suggest why PCR is required in order to identify the presence of TRECs in the blood samples.

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………[2]

(ii) Predict the amount of TRECs present in patients undergoing highly active anti-retroviral therapy (HAART) and give a reason for your answer.

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………[2]

(f) A person who is confirmed as HIV-positive was tested positive for the presence of antibodies to HIV. A human can make more than 10^{12} different antibody molecules as a means to defend against potential pathogens such as the HIV.

Explain how a large diversity of antibodies with different specificity is produced.

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………

………………………………………………………………………………………………[4]

In July 1990, a young woman in Florida, who had no known risk factors for HIV infection and no known contact with other HIV-positive persons, was tested HIV positive after undergoing an invasive dental procedure performed by a dentist who had AIDS.

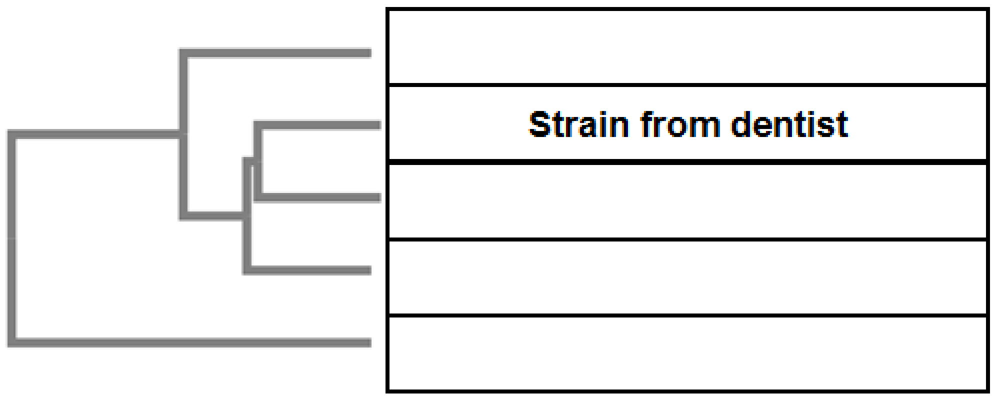
The U.S. Centre for Disease Control and Prevention (CDC) carried out an epidemiological investigation using DNA isolated from white blood cells of the dentist and patients **A** to **D**. The DNA was sequenced and compared with the strain isolated from the dentist.

Fig. 1.3 shows the multiple sequence alignment before a phylogenetic analysis could be carried out.

Strain from	Sequence																			
Dentist	-	-	-	-	C	-	T	A	-	T	T	G	-	C	T	G	G	C	G	C
Patient A	-	-	G	-	C	-	C	A	-	T	A	G	-	C	T	A	G	C	G	C
Patient B	-	-	G	-	C	A	C	C	-	T	-	G	-	C	T	A	G	C	G	C
Patient C	-	-	G	-	C	-	T	-	-	T	G	G	G	C	T	G	G	C	G	C
Patient D	C	A	G	A	C	-	T	A	C	T	-	G	-	C	T	A	G	-	G	-

Fig. 1.3

(h) **(i)** Complete the figure below to show how closely related the different strains of HIV are.



[2]

(ii) Explain how you derive your answer in **(h) (i)**.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………[2]

Viruses are susceptible to the forces of natural selection. Evolutionary changes in the pathogenesis of the HIV have been instrumental in the transmissibility of the virus among people.

Fig. 1.4a and Fig. 1.4b below show the general trend of HIV pathogenesis against the transmissibility of HIV, and the number of infected individuals over time, respectively.

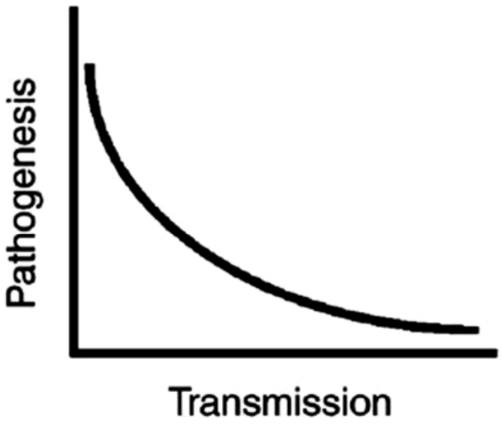


Fig. 1.4a

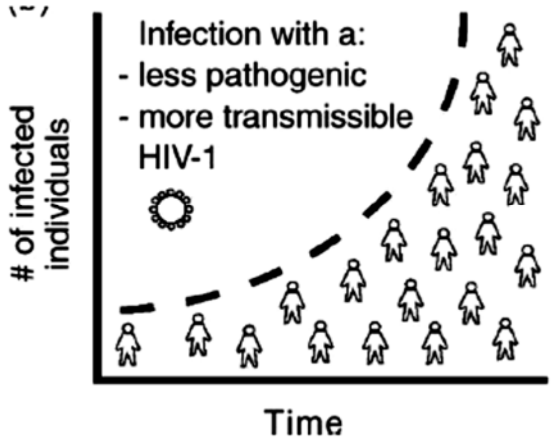


Fig. 1.4b

(i) Explain how the observed trends in Fig. 1.4(a) and (b) support Darwin's theory of natural selection.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………[4]

BLANK PAGE

Bacteria has a natural defence system against viral infections involving RNA sequences known as clustered regularly interspaced short palindromic repeats (CRISPR).

When the bacteriophage infects a bacterial cell, the viral genome released into the bacterial cell is cleaved. Subsequently, a cleaved portion of the viral genome is integrated into the bacterial genome. The bacterial cell detects the phage DNA integrated within its genome and produces a type of RNA known as the CRISPR RNA. The CRISPR RNA contains a sequence that is complementary to that of the integrated phage DNA. When the next phage infects the same bacterial cell, the CRISPR RNA will bind to its target sequence in the viral genome and a nuclease is recruited to cut the phage DNA, disabling the invading phage.

The sequence of the CRISPR RNA can be edited and subsequently used to cut any DNA sequence at a precisely chosen site in eukaryotic cells. This gives rise to the possibility of correcting mutations associated with genetic disorders. Fig. 1.5 shows how the sequence of a defective allele can be replaced with the sequence of a normal allele from a donor DNA via homologous recombination using the CRISPR technology.

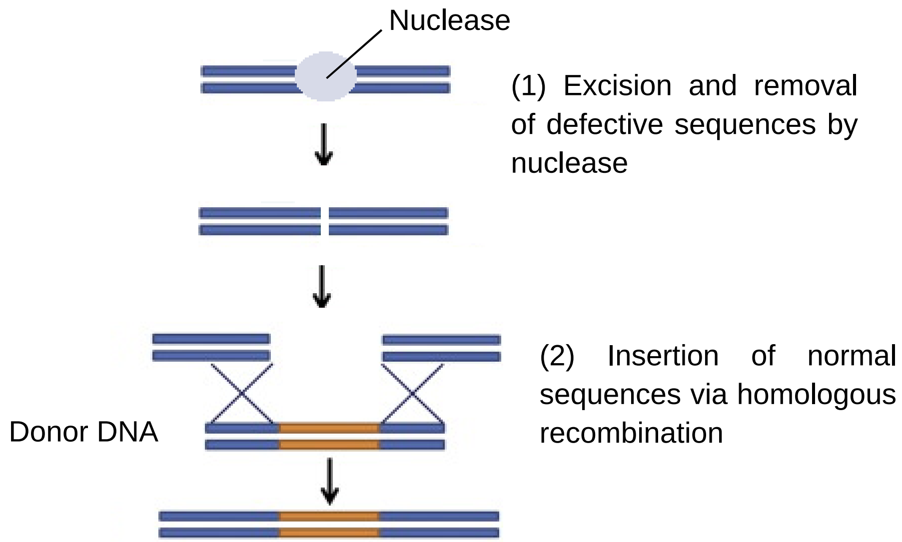


Fig. 1.5

(j) **(i)** Another technology that can be used to treat genetic disorders is gene therapy. Similar to the CRISPR technology, gene therapy involves the introduction of normal alleles into patients. However, it does not involve the excision and removal of defective alleles. Hence, it can only be used to treat recessive genetic disorders.

With reference to Fig. 1.5, explain the advantages of using CRISPR technology to treat genetic disorders over gene therapy.

……………………………………………………………………………………………...

……………………………………………………………………………………………...

……………………………………………………………………………………………...

……………………………………………………………………………………………[2]

(ii) The successful late transcription of HIV provirus following viral activation is highly dependent on the early expression of the viral regulatory proteins Tat and Rev. The elongation of viral mRNA from the integrated provirus is initiated by Tat, while the nuclear export of viral mRNA transcripts is regulated by Rev. In HIV-infected activated T lymphocytes, the combination of Tat and Rev provide a very high level of viral gene expression, while the same proteins in resting T lymphocytes are important for maintaining the provirus in a latent state.

The CRISPR system has been shown in research to abolish Tat and Rev protein expression.

Suggest how the CRISPR system can potentially be used to suppress viral replication.

…………………………………………………………………………………………………..

…………………………………………………………………………………………………[1]

[Total: 30]

2 An investigation was carried out into the response of sorghum being kept at a low temperature for a short period of time. Soybean plants, which are better adapted than sorghum for growth in subtropical and temperate climates, were used for comparison.

Plants of sorghum and soybean were kept at 25°C for several weeks and then at 10°C for three days. The temperature was then increased to 25°C again for seven days. Day length, light intensity and carbon dioxide concentration were kept constant throughout.

The uptake of carbon dioxide, as mg CO_2 absorbed per gram of leaf dry mass, was measured

- at 25 °C before cooling;
- on each of the three days at 10°C;
- for seven days at 25°C. The results are shown in Table 2.1.

Table 2.1

plant	carbon dioxide uptake / mg CO_2 g^{-1}				
	at 25 °C, before cooling	at 10 °C			at 25 °C (mean over days 4 to 10)
		day 1	day 2	day 3	
sorghum	48.2	5.5	2.9	1.2	1.5
soybean	23.2	5.2	3.1	1.6	6.4

(a) Compare the changes in carbon dioxide uptake in sorghum and soybean during the three days at 10°C.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………[2]

During the cooling period, the ultrastructure of the sorghum chloroplasts changed. The membranes of the thylakoids moved closer together, eliminating the spaces between them. The size and number of grana became reduced.

(b) Explain how these changes could be responsible for the low rate of carbon dioxide uptake by sorghum plants even when they are returned to a temperature of 25°C.

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

…………………………………………………………………………………………………….

……………………………………………………………………………………………….[4]

Fig. 2.1 shows the changes in concentration of a 3C compound, glycerate phosphate, GP, and a 5C compound, ribulose bisphosphate, RuBP, extracted from samples taken from actively photosynthesising green algae in an experimental chamber when the light source was turned off.

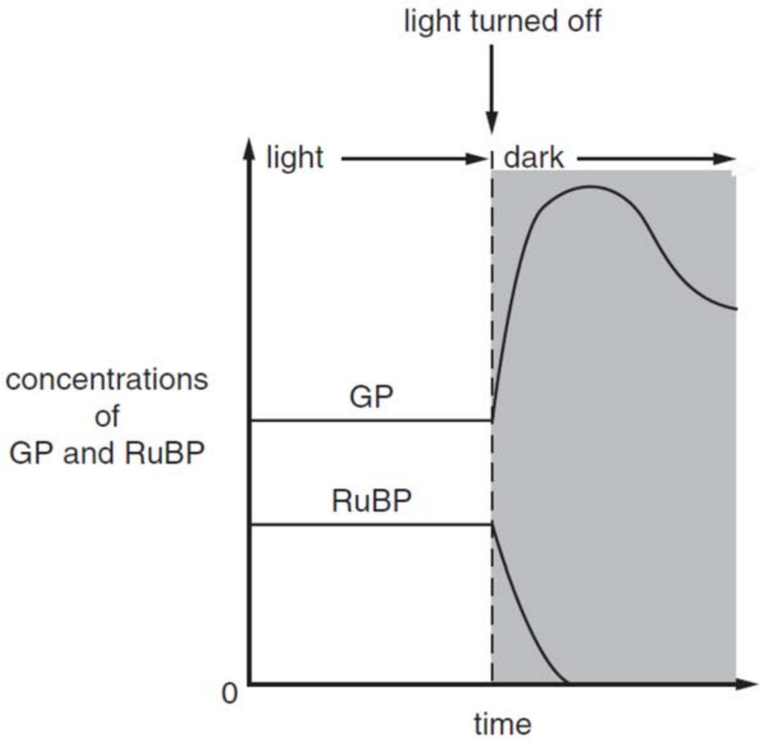


Fig. 2.1

(c) With reference to Fig. 2.1, explain what happens to the concentration of GP and RuBP after the light source was turned off.

……………………………………………………………………………………………………….

……………………………………………………………………………………………………….

……………………………………………………………………………………………………….

……………………………………………………………………………………………………….

……………………………………………………………………………………………………….

……………………………………………………………………………………………………….

……………………………………………………………………………………………………….

…………………………………………………………………………………………………….[4]

[Total: 10]

BLANK PAGE

3 Dengue fever is an infectious disease transmitted by a vector.

(a) State the full name of the vector that transmits the pathogen.

……………………………………………………………………………………………[1]

(b) Global warming could lead to transmission of dengue fever beyond the tropics.

Outline how global warming could change the area of distribution of dengue fever.

…………………………………………………………………………………………….

…………………………………………………………………………………………….

…………………………………………………………………………………………….

……………………………………………………………………………………………[2]

In 2018, the World Health Organization (WHO) estimated that 3.9 billion people were at risk of dengue fever.

Table 3.1 shows the number of cases of dengue fever and the number of deaths from dengue fever between 2000 to 2012, for all the countries of Asia and for Singapore.

The table also shows the numbers in Singapore as percentages of the numbers for all countries in Asia.

Table 3.1

Year	**Number of cases of dengue fever[1]**		**Cases in Singapore as a percentage of Asia**	**Number of deaths from dengue fever[1]**		**Deaths in Singapore as a percentage of Asia**
	Asia	**Singapore**		**Asia**	**Singapore**	
2012	353 210	4 632	1.31	1 248	2	0.16
2010	352 107	5 364	1.52	1 074	6	0.56
2008	206 493	7 032	3.41	668	0	0.00
2006	171 579	3 127	1.82	694	10	1.44
2004	158 848	9 459	5.95	568	8	1.41
2002	108 482	3 945	3.64	514	6	1.17
2000	46 562	673	1.45	247	2	0.81

1 World Health Organisation (WHO) Western Pacific Region

(c) Describe the trends shown in Table 3.1.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

…………………………………………………………………………………………………[4]

(c) Suggest reasons why the number of cases of dengue fever and the number of deaths from dengue fever changed between 2008 and 2012 in Singapore.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

…………………………………………………………………………………………………[3]

[Total: 10]

Section B

Answer **one** question in this section.

Write your answers on separate writing papers provided.

Your answers should be illustrated in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

4 **(a)** Describe how the diversity of bacteria and virus is achieved. [15]

(b) The structural genes in the bacterial operon can be regulated by either the inducible or repressible system.

Discuss the extent, to which the two systems of regulation can be similar and yet different. [10]

[Total: 25]

5 **(a)** Haematopoietic stem cells are found in the bone marrow.

Discuss how haematopoietic stem cells can give rise to naive B-cells which upon activation, will form plasma cells that regulate the production of immunoglobulins to overcome pathogen infection. [15]

(b) Describe the various bonds and their importance in carbohydrates. [10]

[Total: 25]